

An example of relations on the Ext-quiver for the Suzuki group $Sz(8)$ in characteristic 2

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Received 11 December 2004; accepted 28 December 2006

Available online 7 January 2007

Abstract

Let G be the smallest Suzuki group $Sz(8)$ and let F be an algebraically closed field of characteristic 2. The basic algebra of the group algebra of G over F is described by its Ext-quiver and a certain set of relations.

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Keywords: $Sz(8)$; Ext-quiver

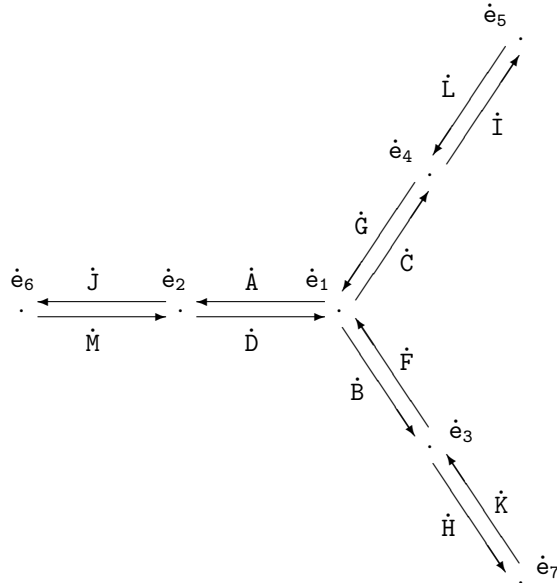
0. Introduction

For a finite-dimensional algebra over a field there is a basic algebra which is Morita equivalent to it (Benson, 1991, Section 2.2). A finite-dimensional basic algebra over an algebraically closed field is described by its Ext-quiver with relations (Benson, 1991, Definitions 4.1.6 and 4.1.8). There is a surjective map of algebras from the path algebra such that the kernel is contained in the ideal of paths of length at least two (Benson, 1991, Proposition 4.1.7). Note that our Ext-quiver has been determined by the simple modules and their extensions.

This paper consists of Introduction, Sections 1–8 and References. Section 1 contains the main theorem and the comments. Section 2 mainly describes the already known valuable results from Schneider (1993) and Sin (1992). Section 3 defines the projective indecomposable modules of larger dimensions and the homomorphisms between the projective indecomposable modules. Section 4 begins by setting an orthogonal primitive idempotent decomposition, defines the homomorphism from the path algebra to the basic algebra which maps the arrows in the quiver

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Fig. 1. The quiver Q .

to the elements corresponding to the homomorphisms between the principal indecomposable modules, and also describes the remark and the lemma used in the succeeding section. Section 5 shows that the previously defined algebra homomorphism is surjective and its kernel is generated by the relations described in the main theorem. Sections 6–8 are for the listings of the GAP programs and the Opal script.

1. Main result

Let G be the Suzuki Group $Sz(8)$, let F be an algebraically closed field of characteristic 2, and let FG be the group algebra of G over F . The group algebra FG has two blocks. Let B_0 be the principal block and B_1 be the other. Then B_1 is a semi-simple algebra containing the Steinberg module.

Theorem 1. Let Q be the quiver as shown in Fig. 1 and let FQ be its path algebra over F . Let $\alpha_1 = \dot{D}\dot{A}$, $\alpha_2 = \dot{F}\dot{B}$, $\alpha_3 = \dot{G}\dot{C}$, $\alpha_4 = \dot{K}\dot{H}$, $\alpha_5 = \dot{L}\dot{I}$, $\alpha_6 = \dot{M}\dot{J}$, $\alpha_7 = \dot{D}\dot{B}\dot{H}$, $\alpha_8 = \dot{F}\dot{C}\dot{I}$, $\alpha_9 = \dot{G}\dot{A}\dot{J}$, $\alpha_{10} = \dot{K}\dot{F}\dot{A}$, $\alpha_{11} = \dot{L}\dot{G}\dot{B}$, $\alpha_{12} = \dot{M}\dot{D}\dot{C}$, $\alpha_{13} = \dot{K}\dot{F}\dot{C}\dot{G}\dot{B}\dot{H}$, $\alpha_{14} = \dot{L}\dot{G}\dot{A}\dot{D}\dot{C}\dot{I}$, $\alpha_{15} = \dot{M}\dot{D}\dot{B}\dot{F}\dot{A}\dot{J}$, $\alpha_{16} = \dot{D}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M} + \dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{A}$, $\alpha_{17} = \dot{F}\dot{A}\dot{D}\dot{B} + \dot{F}\dot{C}\dot{G}\dot{B}\dot{H}\dot{K} + \dot{H}\dot{K}\dot{F}\dot{C}\dot{G}\dot{B}$, $\alpha_{18} = \dot{G}\dot{B}\dot{F}\dot{C} + \dot{G}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L} + \dot{I}\dot{L}\dot{G}\dot{A}\dot{D}\dot{C}$, $\alpha_{19} = \dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{C}\dot{G}\dot{B}\dot{F}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L}\dot{G}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{F}\dot{A}$ $+ \dot{D}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{C}\dot{G}\dot{B}\dot{H}\dot{K}\dot{F}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{C}\dot{I}\dot{L}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}$, $\alpha_{20} = \dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B} + \dot{F}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B} + \dot{F}\dot{C}\dot{G}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B} + \dot{F}\dot{C}\dot{G}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M}\dot{D}\dot{B} + \dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{B}$ $+ \dot{F}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B} + \dot{F}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L}\dot{G}\dot{A}\dot{D}\dot{B} + \dot{F}\dot{A}\dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{B} + \dot{F}\dot{C}\dot{G}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B}$, $\alpha_{21} = \dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C} + \dot{G}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C} + \dot{G}\dot{A}\dot{D}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C} + \dot{G}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{H}\dot{K}\dot{F}\dot{C} + \dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{C}$ $+ \dot{G}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C} + \dot{G}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{C} + \dot{G}\dot{B}\dot{H}\dot{K}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{C} + \dot{G}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{F}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}$, $\beta_1 = \dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G} + \dot{C}\dot{G}\dot{B}\dot{F}\dot{A}\dot{D} + \dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{H}\dot{K}\dot{F}\dot{C}\dot{G} + \dot{B}\dot{H}\dot{K}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}\dot{C} + \dot{C}\dot{G}\dot{A}\dot{D}\dot{C}\dot{G}\dot{B}\dot{H}\dot{K}\dot{F}$ $+ \dot{C}\dot{G}\dot{B}\dot{H}\dot{K}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D} + \dot{A}\dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G} + \dot{C}\dot{G}\dot{B}\dot{F}\dot{A}\dot{D}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M}\dot{D} + \dot{A}\dot{D}\dot{B}\dot{F}\dot{A}\dot{D}\dot{C}\dot{G}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L}\dot{G}$ $+ \dot{C}\dot{G}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{A}\dot{D} + \dot{C}\dot{G}\dot{A}\dot{D}\dot{C}\dot{I}\dot{L}\dot{G}\dot{A}\dot{D}\dot{B}\dot{F}\dot{C}\dot{G}\dot{A}\dot{D}$.

$\alpha_{22} = \beta_1 + \beta_2$ and $\alpha_{23} = \beta_2 + \beta_3$. Let X be the two-sided ideal of FQ generated by $\alpha_1, \alpha_2, \dots, \alpha_{23}$. Then the quotient algebra FQ/X is isomorphic to the basic algebra of B_0 .

2. Preliminaries

Definition 2. Let $i_1, i_2, \dots, i_n$ be integers in $\{0, 1, 2\}$. Denote by $\tau_{(i_1, i_2, \dots, i_n)}$ the group homomorphism from G to the group of nonsingular $4^n \times 4^n$ matrices over F such that

where the symbol $\otimes$ means the Kronecker product of matrices. Note that τ^0 is the identity mapping on G . Let $V_{(i_1, i_2, \dots, i_n)}$ be a 4^n -dimensional row vector space over F . The action of $g \in G$ on $V_{(i_1, i_2, \dots, i_n)}$ is taken to be the standard product of a row vector with the matrix $\tau_{(i_1, i_2, \dots, i_n)}(g)$. Then $V_{(i_1, i_2, \dots, i_n)}$ and $V_{(i_1)} \otimes V_{(i_2)} \otimes \dots \otimes V_{(i_n)}$ are isomorphic FG -modules. Let $V_\emptyset$ be the trivial module F .

Definition 3. Let $P_1, P_2, P_3, P_4, P_5, P_6$ and P_7 be the projective covers of $V_\emptyset, V_{(0)}, V_{(1)}, V_{(2)}, V_{(1,2)}, V_{(0,2)}$ and $V_{(0,1)}$, respectively.

Then we have the next proposition. This is described in [Schneider \(1993, Proposition 6\)](#).

¹ Change the directory in (2) if necessary. The evaluation took about an hour and a half when the author tried.

Proposition 4. $V_{(0,0,1,2)} \cong P_5$, $V_{(0,1,1,2)} \cong P_6$, $V_{(0,1,2,2)} \cong P_7$,
 $V_{(0,1,1,2,2)} \cong P_2 \oplus 2 \cdot V_{(0,1,2)}$, $V_{(0,0,1,2,2)} \cong P_3 \oplus 2 \cdot V_{(0,1,2)}$,
 $V_{(0,0,1,1,2)} \cong P_4 \oplus 2 \cdot V_{(0,1,2)}$, $V_{(0,0,1,1,2,2)} \cong P_1 \oplus 2 \cdot P_5 \oplus 2 \cdot P_6 \oplus 2 \cdot P_7 \oplus 9 \cdot V_{(0,1,2)}$.

Corollary 5. $\dim_F P_5 = \dim_F P_6 = \dim_F P_7 = 256$,
 $\dim_F P_2 = \dim_F P_3 = \dim_F P_4 = 896$, $\dim_F P_1 = 1984$.

3. Row vector spaces

We are going to use p.l.g. Note that $r_1 = \tau(r_0)$ by (14) and $r_2 = \tau^2(r_0)$ by (15), while $\tau(s_0) = s_0$ by (10)–(13). For example, (16) computes $r_0 \otimes r_0 \otimes r_1 \otimes r_2 = \tau_{(0,0,1,2)}(r_0)$ and binds it to r_3 .

Definition 6. For a matrix M over F , let $V(M)$ be a row vector space over F spanned by the whole rows of M . Note that an element of $V(M)$ is a standard product of a row vector with M .

The matrix e_0 commutes with both $\tau_{(0,0,1,1,2,2)}(r_0)$ and $\tau_{(0,0,1,1,2,2)}(s_0)$ by (24), (25) and (29). The group G is generated by $\{r_0, s_0\}$. Hence e_0 commutes with $\tau_{(0,0,1,1,2,2)}(g)$ for all $g \in G$. Thus $V(e_0)$ is an FG -submodule of $V_{(0,0,1,1,2,2)}$. Similarly, by (20)–(25) and (30)–(33), $V(e_1)$, $V(e_2)$, $V(e_3)$ and $V(e_4)$ are FG -submodules of $V_{(0,0,1,1,2,2)}$, $V_{(0,1,1,2,2)}$, $V_{(0,0,1,2,2)}$ and $V_{(0,0,1,1,2)}$, respectively.

Lemma 7. $V(e_1) \cong P_1$, $V(e_2) \cong P_2$, $V(e_3) \cong P_3$, $V(e_4) \cong P_4$.

Proof. Since $(e_0)^2 = e_0$ by (34), the module $V(e_0)$ is a direct summand of $V_{(0,0,1,1,2,2)}$. Similarly, by (35)–(38), the modules $V(e_1)$, $V(e_2)$, $V(e_3)$ and $V(e_4)$ are direct summands of $V_{(0,0,1,1,2,2)}$, $V_{(0,1,1,2,2)}$, $V_{(0,0,1,2,2)}$ and $V_{(0,0,1,1,2)}$, respectively. By (41)–(43), $\dim_F V(e_2) = \dim_F V(e_3) = \dim_F V(e_4) = 896$. Thus Proposition 4 and the Krull–Schmidt theorem yield $V(e_2) \cong P_2$, $V(e_3) \cong P_3$ and $V(e_4) \cong P_4$. Since both $e_0 \cdot e_1$ and $e_1 \cdot e_0$ are zero matrices by (44), $V(e_0) \oplus V(e_1)$ is a direct summand of $V_{(0,0,1,1,2,2)}$. We have

$$\begin{aligned} \dim_F V(e_0) &= 576 < \dim_F P_1, \\ \dim_F V(e_1) &= 1984 = \dim_F P_1, \\ \dim_F V_{(0,0,1,1,2,2)} - \dim_F (V(e_0) \oplus V(e_1)) &= 4096 - (576 + 1984) < \dim_F P_1 \end{aligned}$$

by (39), (40) and Corollary 5. Thus $V(e_1) \cong P_1$ by Proposition 4 and the Krull–Schmidt theorem. $\square$

Since $A \cdot e_1 = A$ by (45), the standard product $x \cdot A$ of a row vector $x \in V(e_2)$ with the matrix A belongs to $V(e_1)$. By (20), (23)–(25) and (54), the equation $\tau_{(0,1,1,2,2)}(g) \cdot A = A \cdot \tau_{(0,0,1,1,2,2)}(g)$ holds for $g = r_0, s_0$. The group G is generated by $\{r_0, s_0\}$. Hence the same equation holds for all $g \in G$. Thus the homomorphism which maps $x \in V(e_2)$ to $x \cdot A \in V(e_1)$ is an FG -homomorphism. We use (16)–(25) and (45)–(65) in this manner.

Definition 8. FG -homomorphisms

$$\begin{array}{lll} \bar{a} : V(e_2) \rightarrow V(e_1), & \bar{b} : V(e_3) \rightarrow V(e_1), & \bar{c} : V(e_4) \rightarrow V(e_1), \\ \bar{d} : V(e_1) \rightarrow V(e_2), & \bar{f} : V(e_1) \rightarrow V(e_3), & \bar{g} : V(e_1) \rightarrow V(e_4), \\ \bar{h} : V_{(0,1,2,2)} \rightarrow V(e_3), & \bar{i} : V_{(0,0,1,2)} \rightarrow V(e_4), & \bar{j} : V_{(0,1,1,2)} \rightarrow V(e_2), \\ \bar{k} : V(e_3) \rightarrow V_{(0,1,2,2)}, & \bar{l} : V(e_4) \rightarrow V_{(0,0,1,2)}, & \bar{m} : V(e_2) \rightarrow V_{(0,1,1,2)} \end{array}$$

can be defined by

$$\begin{array}{lll} \bar{a}(x) = x \cdot A, & \bar{b}(x) = x \cdot B, & \bar{c}(x) = x \cdot C, \\ \bar{d}(x) = x \cdot D, & \bar{f}(x) = x \cdot F, & \bar{g}(x) = x \cdot G, \\ \bar{h}(x) = x \cdot H, & \bar{i}(x) = x \cdot I, & \bar{j}(x) = x \cdot J, \\ \bar{k}(x) = x \cdot K, & \bar{l}(x) = x \cdot L, & \bar{m}(x) = x \cdot M, \end{array}$$

where the right hand sides are standard products of a row vector with a matrix.

4. Idempotent decomposition

Definition 9. Let $A = FG$ and let $1_A = \varepsilon_0 + \varepsilon_1$ where ε_0 and ε_1 are central idempotents such that $B_0 = \varepsilon_0 A$ and $B_1 = \varepsilon_1 A$. Let

$$\varepsilon_0 = e_1 + \sum_{a=1}^4 e_{2,a} + \sum_{a=1}^4 e_{3,a} + \sum_{a=1}^4 e_{4,a} + \sum_{a=1}^{16} e_{5,a} + \sum_{a=1}^{16} e_{6,a} + \sum_{a=1}^{16} e_{7,a}$$

be an orthogonal primitive idempotent decomposition of ε_0 such that $e_1 A \cong P_1$ and $e_{2,a} A \cong P_2$, $e_{3,a} A \cong P_3$, $e_{4,a} A \cong P_4$ (for $a = 1, 2, 3, 4$) and $e_{5,a} A \cong P_5$, $e_{6,a} A \cong P_6$, $e_{7,a} A \cong P_7$ (for $a = 1, \dots, 16$). Let $e_i = e_{i,1}$ for $i = 2, \dots, 7$.

Definition 10. We fix the following A -isomorphisms. (Proposition 4 gives the latter three. Lemma 7 gives the others.)

$$\begin{array}{lll} \eta_1 : V(e_1) \rightarrow e_1 A, & \eta_2 : V(e_2) \rightarrow e_2 A, & \eta_3 : V(e_3) \rightarrow e_3 A, \\ \eta_4 : V(e_4) \rightarrow e_4 A, & \eta_5 : V_{(0,0,1,2)} \rightarrow e_5 A, & \eta_6 : V_{(0,1,1,2)} \rightarrow e_6 A, \\ \eta_7 : V_{(0,1,2,2)} \rightarrow e_7 A. \end{array}$$

Definition 11. We define the following A -homomorphisms.

$$\begin{array}{ll} \tilde{a} = \eta_1 \circ \bar{a} \circ (\eta_2)^{-1} : e_2 A \rightarrow e_1 A, & \tilde{b} = \eta_1 \circ \bar{b} \circ (\eta_3)^{-1} : e_3 A \rightarrow e_1 A, \\ \tilde{c} = \eta_1 \circ \bar{c} \circ (\eta_4)^{-1} : e_4 A \rightarrow e_1 A, & \tilde{d} = \eta_2 \circ \bar{d} \circ (\eta_1)^{-1} : e_1 A \rightarrow e_2 A, \\ \tilde{f} = \eta_3 \circ \bar{f} \circ (\eta_1)^{-1} : e_1 A \rightarrow e_3 A, & \tilde{g} = \eta_4 \circ \bar{g} \circ (\eta_1)^{-1} : e_1 A \rightarrow e_4 A, \\ \tilde{h} = \eta_3 \circ \bar{h} \circ (\eta_7)^{-1} : e_7 A \rightarrow e_3 A, & \tilde{i} = \eta_4 \circ \bar{i} \circ (\eta_5)^{-1} : e_5 A \rightarrow e_4 A, \\ \tilde{j} = \eta_2 \circ \bar{j} \circ (\eta_6)^{-1} : e_6 A \rightarrow e_2 A, & \tilde{k} = \eta_7 \circ \bar{k} \circ (\eta_3)^{-1} : e_3 A \rightarrow e_7 A, \\ \tilde{l} = \eta_5 \circ \bar{l} \circ (\eta_4)^{-1} : e_4 A \rightarrow e_5 A, & \tilde{m} = \eta_6 \circ \bar{m} \circ (\eta_2)^{-1} : e_2 A \rightarrow e_6 A. \end{array}$$

Let $e = e_1 + e_2 + \dots + e_7$ and let $\Lambda = eAe (= eB_0e)$. Then Λ is a basic algebra of B_0 . Let $J(A)$ be the Jacobson radical of A and let $J = eJ(A)e$. Then J is the Jacobson radical of Λ (Feit, 1982, Lemma I.8.4). Note that $J \subset J(A)$. We have $e_i A e_j = e_i A e_j$, $e_i J e_j = e_i J(A) e_j$ and

$$e_i A e_j / e_i J(A) e_j \cong \text{Hom}_A(e_j A, e_i A / e_i J(A)) \cong \begin{cases} F & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

for all $i, j = 1, \dots, 7$ (Benson, 1991, Lemmas 1.3.3(i) and 1.7.5).

Definition 12. We define the following elements of Λ .

$$\begin{array}{lll} \underline{a} = \tilde{a}(e_2) \in e_1 J e_2, & \underline{b} = \tilde{b}(e_3) \in e_1 J e_3, & \underline{c} = \tilde{c}(e_4) \in e_1 J e_4, \\ \underline{d} = \tilde{d}(e_1) \in e_2 J e_1, & \underline{f} = \tilde{f}(e_1) \in e_3 J e_1, & \underline{g} = \tilde{g}(e_1) \in e_4 J e_1, \\ \underline{h} = \tilde{h}(e_7) \in e_3 J e_7, & \underline{i} = \tilde{i}(e_5) \in e_4 J e_5, & \underline{j} = \tilde{j}(e_6) \in e_2 J e_6, \\ \underline{k} = \tilde{k}(e_3) \in e_7 J e_3, & \underline{l} = \tilde{l}(e_4) \in e_5 J e_4, & \underline{m} = \tilde{m}(e_2) \in e_6 J e_2. \end{array}$$

Definition 13. Let ϕ be the F -algebra homomorphism from FQ to Λ such that

$$\begin{array}{llllll} \phi(\dot{e}_1) = e_1, & \phi(\dot{e}_2) = e_2, & \phi(\dot{e}_3) = e_3, & \phi(\dot{e}_4) = e_4, & \phi(\dot{e}_5) = e_5, \\ \phi(\dot{e}_6) = e_6, & \phi(\dot{e}_7) = e_7, & \phi(\dot{A}) = \underline{a}, & \phi(\dot{B}) = \underline{b}, & \phi(\dot{C}) = \underline{c}, \\ \phi(\dot{D}) = \underline{d}, & \phi(\dot{F}) = \underline{f}, & \phi(\dot{G}) = \underline{g}, & \phi(\dot{H}) = \underline{h}, & \phi(\dot{I}) = \underline{i}, \\ \phi(\dot{J}) = \underline{j}, & \phi(\dot{K}) = \underline{k}, & \phi(\dot{L}) = \underline{l}, & \phi(\dot{M}) = \underline{m}. \end{array}$$

Calculations such as $\underline{a} \eta_2(x_2) = \tilde{a}(e_2) \eta_2(x_2) = \tilde{a}(e_2 \eta_2(x_2)) = \tilde{a}(\eta_2(x_2)) = \eta_1(\bar{a}(x_2)) = \eta_1(x_2 \cdot A)$ ($\forall x_2 \in V(e_2)$) yield the next remark.

Remark 14. For any $x_1 \in V(e_1)$, $x_2 \in V(e_2)$, $x_3 \in V(e_3)$, $x_4 \in V(e_4)$, $x_5 \in V_{(0,0,1,2)}$, $x_6 \in V_{(0,1,1,2)}$ and $x_7 \in V_{(0,1,2,2)}$, the following equations hold.

$$\begin{array}{lll} \underline{a} \eta_2(x_2) = \eta_1(x_2 \cdot A), & \underline{b} \eta_3(x_3) = \eta_1(x_3 \cdot B), & \underline{c} \eta_4(x_4) = \eta_1(x_4 \cdot C), \\ \underline{d} \eta_1(x_1) = \eta_2(x_1 \cdot D), & \underline{f} \eta_1(x_1) = \eta_3(x_1 \cdot F), & \underline{g} \eta_1(x_1) = \eta_4(x_1 \cdot G), \\ \underline{h} \eta_7(x_7) = \eta_3(x_7 \cdot H), & \underline{i} \eta_5(x_5) = \eta_4(x_5 \cdot I), & \underline{j} \eta_6(x_6) = \eta_2(x_6 \cdot J), \\ \underline{k} \eta_3(x_3) = \eta_7(x_3 \cdot K), & \underline{l} \eta_4(x_4) = \eta_5(x_4 \cdot L), & \underline{m} \eta_2(x_2) = \eta_6(x_2 \cdot M). \end{array}$$

Lemma 15. Let $\mathfrak{t}_1, \dots, \mathfrak{t}_7$ be the elements defined in p2.g.

- (i) $V(e_1) = \mathfrak{t}_1 \cdot A$, $V(e_2) = \mathfrak{t}_2 \cdot A$, $V(e_3) = \mathfrak{t}_3 \cdot A$, $V(e_4) = \mathfrak{t}_4 \cdot A$, $V_{(0,0,1,2)} = \mathfrak{t}_5 \cdot A$,
 $V_{(0,1,1,2)} = \mathfrak{t}_6 \cdot A$, $V_{(0,1,2,2)} = \mathfrak{t}_7 \cdot A$.
- (ii) $\underline{a}, \underline{b}, \underline{c}, \underline{d}, \underline{f}, \underline{g}, \underline{h}, \underline{i}, \underline{j}, \underline{k}, \underline{l}, \underline{m} \notin J^2$.

Proof. First note that $\mathfrak{t}_1 \in V(e_1)$ by (66). By (70) and (71),

$$\mathfrak{t}_1 \cdot F \cdot B \cdot D \cdot M \cdot J \cdot A \cdot F \cdot B \cdot D \cdot A \cdot G \cdot L \cdot I \cdot C \cdot D \cdot A \cdot F \cdot B \cdot D \cdot A \cdot G \cdot L \cdot I \cdot C \cdot D \cdot A \neq 0.$$

Thus

$$\begin{aligned} & \underline{a} \underline{d} \underline{c} \underline{i} \underline{l} \underline{g} \underline{a} \underline{d} \underline{b} \underline{f} \underline{a} \underline{d} \underline{c} \underline{i} \underline{l} \underline{g} \underline{a} \underline{d} \underline{b} \underline{f} \underline{a} \underline{j} \underline{m} \underline{d} \underline{b} \underline{f} \eta_1(\mathfrak{t}_1) \\ &= \eta_1(\mathfrak{t}_1 \cdot F \cdot B \cdot D \cdot M \cdot J \cdot A \cdot F \cdot B \cdot D \cdot A \cdot G \cdot L \cdot I \cdot C \cdot D \cdot A \cdot F \cdot B \cdot D \cdot A \cdot G \cdot L \cdot I \cdot C \cdot D \cdot A) \neq 0 \end{aligned}$$

by using Remark 14 repeatedly. By Schneider (1993, Table 4), $e_1 A$ has socle length 27. This is equal to the radical length (Benson, 1991, Definition 1.2.1). Hence $e_1 J(A)^{27} = 0$. Thus

$$\underline{a} \underline{d} \underline{c} \underline{i} \underline{l} \underline{g} \underline{a} \underline{d} \underline{b} \underline{f} \underline{a} \underline{d} \underline{c} \underline{i} \underline{l} \underline{g} \underline{a} \underline{d} \underline{b} \underline{f} \underline{a} \underline{j} \underline{m} \underline{d} \underline{b} \underline{f} \eta_1(\mathfrak{t}_1) \notin J(A)^{27}.$$

Hence $\underline{a}, \underline{b}, \underline{c}, \underline{d}, \underline{f}, \underline{g}, \underline{i}, \underline{j}, \underline{l}, \underline{m} \notin J(A)^2$ and $\eta_1(\mathfrak{t}_1) \notin e_1 J(A)$ because $\underline{a}, \underline{b}, \underline{c}, \underline{d}, \underline{f}, \underline{g}, \underline{i}, \underline{j}, \underline{l}, \underline{m} \in J(A)$. Then $\eta_1(\mathfrak{t}_1)$ generates $e_1 A$ as A -module because $e_1 J(A)$ is a unique maximal A -submodule of $e_1 A$. Therefore $\mathfrak{t}_1$ generates $V(e_1)$ as A -module. The rest is obtained from Schneider (1993, Tables 2 and 3), (67)–(69) and (72)–(83) in a similar way. $\square$

5. Algebra homomorphism

The proof of [Theorem 1](#) is completed in this section. The basic algebra Λ and the homomorphism ϕ were defined in the previous section. It will turn out that ϕ is surjective and its kernel is X defined in [Theorem 1](#).

Lemma 16. $X \subset \text{Ker } \phi$.

Proof. It suffices to show that $\alpha_i \in \text{Ker } \phi$ for all $i = 1, \dots, 23$. This is achieved by using (84)–(168). We are going to show only $\alpha_{16} = \dot{D}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M} + \dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{A} \in \text{Ker } \phi$. Let y be a row vector in $V(e_2)$ such that $\eta_2(y) = e_2$. By [Lemma 15\(i\)](#), some z in A satisfies $y = t_2 \cdot z$. Then

$$\begin{aligned} \underline{d} \underline{c} \underline{g} \underline{a} &= \underline{d} \underline{c} \underline{g} \underline{a} \eta_2(t_2) z = \eta_2(t_2 \cdot A \cdot G \cdot C \cdot D) z, \\ \underline{d} \underline{b} \underline{f} \underline{a} \underline{j} \underline{m} &= \underline{d} \underline{b} \underline{f} \underline{a} \underline{j} \underline{m} \eta_2(t_2) z = \eta_2(t_2 \cdot M \cdot J \cdot A \cdot F \cdot B \cdot D) z, \\ \underline{j} \underline{m} \underline{d} \underline{b} \underline{f} \underline{a} &= \underline{j} \underline{m} \underline{d} \underline{b} \underline{f} \underline{a} \eta_2(t_2) z = \eta_2(t_2 \cdot A \cdot F \cdot B \cdot D \cdot M \cdot J) z \end{aligned}$$

by [Remark 14](#). We have $t_2 \cdot A \cdot G \cdot C \cdot D + t_2 \cdot M \cdot J \cdot A \cdot F \cdot B \cdot D + t_2 \cdot A \cdot F \cdot B \cdot D \cdot M \cdot J = 0$ by (99)–(101). Thus

$$\begin{aligned} \phi(\dot{D}\dot{C}\dot{G}\dot{A} + \dot{D}\dot{B}\dot{F}\dot{A}\dot{J}\dot{M} + \dot{J}\dot{M}\dot{D}\dot{B}\dot{F}\dot{A}) \\ &= \underline{d} \underline{c} \underline{g} \underline{a} + \underline{d} \underline{b} \underline{f} \underline{a} \underline{j} \underline{m} + \underline{j} \underline{m} \underline{d} \underline{b} \underline{f} \underline{a} \\ &= \eta_2(t_2 \cdot A \cdot G \cdot C \cdot D + t_2 \cdot M \cdot J \cdot A \cdot F \cdot B \cdot D + t_2 \cdot A \cdot F \cdot B \cdot D \cdot M \cdot J) z \\ &= 0. \quad \square \end{aligned}$$

Definition 17. There is a unique F -algebra homomorphism ψ from FQ/X to Λ such that $\phi = \psi \circ \pi$ where π is a canonical surjection from FQ to FQ/X .

Proposition 18. ϕ is surjective.

Proof. Note that $\dim_F(e_i \Lambda e_i / e_i J e_i) = 1$ and $e_i \in e_i \Lambda e_i$ but $e_i \notin e_i J e_i$ for all $i = 1, \dots, 7$, while $e_i \Lambda e_j = e_i J e_j$ for all $i, j = 1, \dots, 7$ such that $i \neq j$. Thus ϕ is surjective modulo J . Let $i, j = 1, \dots, 7$. If $e_i J e_j \neq e_i J^2 e_j$ then $\dim_F(e_i J e_j / e_i J^2 e_j) = 1$ and $e_i J e_j$ contains some element among $\underline{a}, \underline{b}, \underline{c}, \underline{d}, \underline{f}, \underline{g}, \underline{h}, \underline{i}, \underline{j}, \underline{k}, \underline{l}, \underline{m}$ by [Benson \(1991, Proposition 2.4.3\)](#) and [Sin \(1992, Theorem\)](#), but it does not belong to $e_i J^2 e_j$ by [Lemma 15\(ii\)](#). Thus ϕ is surjective modulo J^2 . Hence ϕ is surjective by [Benson \(1991, Proposition 1.2.8\)](#). $\square$

Corollary 19. ψ is surjective.

Proposition 20. ψ is an F -algebra isomorphism.

Proof. For a nonzero z in FQ , let $\text{supp}(z)$ be the set of paths appearing in the linear combination representing z with nonzero coefficients, and let $t(z)$ be the largest path among $\text{supp}(z)$. For a subset Z of FQ , let $t(Z)$ be the set of $t(z)$ where $z \in Z \setminus \{0\}$, and let $u(Z)$ be the set of all paths which are divided by some path from $t(Z)$ (cf. [Farkas et al. \(1993\)](#)). Let $\mathbb{F}_2 = \{0, 1\} \subset F$. Let $\mathbb{F}_2 Q$ be the path algebra of Q over $\mathbb{F}_2$. Let Y be the two-sided ideal of $\mathbb{F}_2 Q$ generated by $\alpha_1, \alpha_2, \dots, \alpha_{23}$. Note that $\mathbb{F}_2 Q \subset FQ$ and $Y \subset X$. We are going to use p3.opal. There is correspondence $\dot{e}_1 \leftrightarrow (1), \dot{e}_2 \leftrightarrow (2), \dots, \dot{e}_7 \leftrightarrow (7), \dot{A} \leftrightarrow A, \dot{B} \leftrightarrow B, \dots, \dot{M} \leftrightarrow M$ between the quiver Q and the graph g of p3.opal. Since `IsGrobner(S)` returns `True`, the identifier S is bound to the Gröbner basis for Y . Thus $u(S) = t(Y)$ (see [Farkas et al. \(1993, p. 731\)](#)). Let Π be the set of all paths in the quiver Q . Then $\Pi \setminus t(X) \subset \Pi \setminus u(S)$ because $t(Y) \subset t(X)$. The function `Nontips(S, 18)` returns the intersection of $\Pi \setminus u(S)$ and the set of paths of length less than or

equal to 18. The function `Nontips(S, 19)` returns the same result in our case. This means that $u(S)$ contains all paths of length 19. Hence $u(S)$ contains all paths of length greater than or equal to 19. Therefore `Nontips(S, 18)` returns $\Pi \setminus u(S)$ itself. The result of `Nontips(S, 18)` is the set of 1192 paths. Thus $|\Pi \setminus t(X)| \leq 1192$. By Farkas et al. (1993, Theorem 4), FQ is a direct sum of X and the F -space spanned by the whole paths in $\Pi \setminus t(X)$. Hence $\dim_F FQ/X \leq 1192$. The sum of all entries in the Cartan matrix for the principal block is equal to $\dim_F \Lambda$. Thus $\dim_F \Lambda = 1192$ by Schneider (1993, Appendix). Since ψ is surjective, it turns out that ψ is bijective. $\square$

6. GAP program p1.g

The function `f5(x, y, z)` defined in (26)–(28) returns the same Boolean value as $x * y = y * z$ but seems to be faster in our case. The following program is p1.g.

```

g1 := Sz(8); (1)
Read("./p2.g"); (2)
ans := []; (3)
g2 := GeneratorsOfGroup(g1); r0 := g2[1]; s0 := g2[2]; (4)
ans[1] := Length(g2) = 2; (5)
ans[2] := r0 = [[Z(2)^0, 0*Z(2), 0*Z(2), 0*Z(2)], (6)
               [Z(2)^0, Z(2)^0, 0*Z(2), 0*Z(2)], (7)
               [Z(8)^3, Z(2)^0, Z(2)^0, 0*Z(2)], (8)
               [Z(8)^6, Z(8)^1, Z(2)^0, Z(2)^0]]; (9)
ans[3] := s0 = [[0*Z(2), 0*Z(2), 0*Z(2), Z(2)^0], (10)
               [0*Z(2), 0*Z(2), Z(2)^0, 0*Z(2)], (11)
               [0*Z(2), Z(2)^0, 0*Z(2), 0*Z(2)], (12)
               [Z(2)^0, 0*Z(2), 0*Z(2), 0*Z(2)]]; (13)
r1 := List(r0, x->List(x, y->y^4)); (14)
r2 := List(r0, x->List(x, y->y^2)); (15)
r3 := Iterated([r0, r0, r1, r2], KroneckerProduct); (16)
r4 := Iterated([r0, r1, r1, r2], KroneckerProduct); (17)
r5 := Iterated([r0, r1, r2, r2], KroneckerProduct); (18)
s1 := Iterated([s0, s0, s0, s0], KroneckerProduct); (19)
r6 := Iterated([r0, r1, r1, r2, r2], KroneckerProduct); (20)
r7 := Iterated([r0, r0, r1, r2, r2], KroneckerProduct); (21)
r8 := Iterated([r0, r0, r1, r1, r2], KroneckerProduct); (22)
s2 := Iterated([s0, s0, s0, s0, s0], KroneckerProduct); (23)
r9 := Iterated([r0, r0, r1, r1, r2, r2], KroneckerProduct); (24)
s3 := Iterated([s0, s0, s0, s0, s0, s0], KroneckerProduct); (25)
f5 := function(x, y, z) (26)
  return TransposedMat(y) * TransposedMat(x) (27)
        = TransposedMat(y * z); end; (28)
ans[4] := f5(r9, e0, r9); ans[5] := f5(s3, e0, s3); (29)
ans[6] := f5(r9, e1, r9); ans[7] := f5(s3, e1, s3); (30)
ans[8] := f5(r6, e2, r6); ans[9] := f5(s2, e2, s2); (31)
ans[10] := f5(r7, e3, r7); ans[11] := f5(s2, e3, s2); (32)
ans[12] := f5(r8, e4, r8); ans[13] := f5(s2, e4, s2); (33)

```


$$+ t4*C*F*B*D*A*G*C*D*A*G \quad (128)$$

$$+ t4*C*F*K*H*B*G*C*D*A*G \quad (129)$$

$$+ t4*C*D*A*G*C*F*B*D*A*G \quad (130)$$

$$+ t4*C*F*B*D*A*G*C*F*B*G \quad (131)$$

$$+ t4*C*F*B*D*M*J*A*F*B*G \quad (132)$$

$$+ t4*C*D*A*G*C*F*K*H*B*G \quad (133)$$

$$+ t4*C*F*B*D*A*F*B*G*C*D*A*G); \quad (134)$$

$$x := t1*G*C*F*B*D*A \quad (135)$$

$$+ t1*D*A*F*B*G*C \quad (136)$$

$$+ t1*G*C*F*K*H*B*G*C*D*A \quad (137)$$

$$+ t1*G*C*D*A*G*C*F*K*H*B \quad (138)$$

$$+ t1*F*K*H*B*G*C*D*A*G*C \quad (139)$$

$$+ t1*D*A*G*C*F*K*H*B*G*C \quad (140)$$

$$+ t1*G*C*F*B*D*A*F*B*D*M*J*A \quad (141)$$

$$+ t1*D*M*J*A*F*B*D*A*F*B*G*C \quad (142)$$

$$+ t1*G*L*I*C*D*A*G*C*D*A*F*B*D*A \quad (143)$$

$$+ t1*D*A*F*B*D*A*G*L*I*C*D*A*G*C \quad (144)$$

$$+ t1*D*A*G*C*F*B*D*A*G*L*I*C*D*A*G*C; \quad (145)$$

$$y := t1*D*A*G*C*F*B \quad (146)$$

$$+ t1*F*B*G*C*D*A \quad (147)$$

$$+ t1*D*A*G*L*I*C*D*A*F*B \quad (148)$$

$$+ t1*D*A*F*B*D*A*G*L*I*C \quad (149)$$

$$+ t1*G*L*I*C*D*A*F*B*D*A \quad (150)$$

$$+ t1*F*B*D*A*G*L*I*C*D*A \quad (151)$$

$$+ t1*D*A*G*C*F*B*G*C*F*K*H*B \quad (152)$$

$$+ t1*F*K*H*B*G*C*F*B*G*C*D*A \quad (153)$$

$$+ t1*D*M*J*A*F*B*D*A*F*B*G*C*F*B \quad (154)$$

$$+ t1*F*B*G*C*F*B*D*M*J*A*F*B*D*A \quad (155)$$

$$+ t1*F*B*D*A*G*C*F*B*D*M*J*A*F*B*D*A; \quad (156)$$

$$z := t1*F*B*D*A*G*C \quad (157)$$

$$+ t1*G*C*D*A*F*B \quad (158)$$

$$+ t1*F*B*D*M*J*A*F*B*G*C \quad (159)$$

$$+ t1*F*B*G*C*F*B*D*M*J*A \quad (160)$$

$$+ t1*D*M*J*A*F*B*G*C*F*B \quad (161)$$

$$+ t1*G*C*F*B*D*M*J*A*F*B \quad (162)$$

$$+ t1*F*B*D*A*G*C*D*A*G*L*I*C \quad (163)$$

$$+ t1*G*L*I*C*D*A*G*C*D*A*F*B \quad (164)$$

$$+ t1*F*K*H*B*G*C*F*B*G*C*D*A*G*C \quad (165)$$

$$+ t1*G*C*D*A*G*C*F*K*H*B*G*C*F*B \quad (166)$$

$$+ t1*G*C*F*B*D*A*G*C*F*K*H*B*G*C*F*B; \quad (167)$$

$$\text{ans}[91] := x = y; \quad \text{ans}[92] := y = z; \quad (168)$$

7. GAP program p2.g

The function $f2$ with the assistance of $f1$ is used to get $\sum_{g \in G_2} \tau_{(0,1,1,2,2)}(g)$ and $\sum_{g \in G_2} \tau_{(0,0,1,1,2,2)}(g)$ where G_2 is the subset of G consisting of all elements of even order. The functions $f3$ and $f4$ are used to get the matrix of which the entries are permuted according

to 4-adic representations of nonnegative integers depending on a position of each entry. The following program is p2.g.

```
f1 := function(x) local a,b,c,d,e,f,g,h,i;
  a := List(x,y->List([1..4^Length(y)],z->[]));
  b := [[1,2],[1,4],[2,1],[4,2]];
  for c in ConjugacyClasses(g1) do
    if IsEvenInt(Order(Representative(c))) then
      for d in AsList(c) do
        e := PositionProperty(b,y->not d[y[1]][y[2]]
                               in [0*Z(2),Z(2)^0]);
        if e = fail or d[b[e][1]][b[e][2]] in [Z(8)^1,Z(8)^3]
        then f := [d,List(d,y->List(y,z->z^4))
                   ,List(d,y->List(y,z->z^2))];
        for g in [1..Length(x)] do
          h := Iterated(List(x[g],y->f[y]),KroneckerProduct);
          for i in [1..Length(h)] do
            Add[a[g][i],h[i]]; od; od; fi; od; fi; od;
          for b in [1..Length(x)] do
            for c in [1..Length(a[b])] do
              TransposedMatDestructive(a[b][c]); od; od;
            return a; end;
          fi; od;
        fi; od;
      fi; od;
    fi; od;
  fi; od;
f2 := function(x,y) local a,b,c,d,e,f,g,h;
  a := Length(x) * Length(y); b := NullMat(a,a,Z(2)); a := 0;
  for c in x do for d in y do a := a + 1; e := 0;
    for f in c do for g in d do e := e + 1;
      h := f * g; b[a][e] := h + h^2 + h^4; od; od; od; od;
    od; od;
  return b; end;
f3 := function(x,y,z) local a,b,c,d,e,f,g,h,i,j,k;
  a := Length(y); b := Length(z); c := 4^a; d := 4^b;
  e := List(y,x->4^x); f := List(z,x->4^x);
  g := NullMat(c,d,Z(2));
  for h in [1..c] do i := g[h];
    j := x[e * CoefficientsQadic(h + c - 1,4){[1..a]} + 1];
    for k in [1..d] do
      i[k] := j[f * CoefficientsQadic(k + d - 1,4){[1..b]} + 1];
    od; od; return g; end;
f4 := function(x,y) return f3(x,y,y); end;
x := f1([1,2],[1,1,2],[2,3,3]); e2 := f2(x[1],x[3]);
e3 := f4(e2,[2,3,4,0,1]); e4 := f4(e2,[4,0,1,2,3]);
e0 := f2(x[2],x[3]); e0 := e0^2 + IdentityMat(4096,Z(2));
h0 := IdentityMat(4,Z(2)); h1 := KroneckerProduct(h0,e2);
h2 := f4(h1,[2,3,4,5,0,1]); h3 := f4(h1,[4,5,0,1,2,3]);
e1 := (e0 + h1 + h2 + h3)^4; e5 := IdentityMat(256,Z(2));
t1 := e1[820]; t2 := e2[52]; t3 := e3[196]; t4 := e4[205];
t5 := e5[49]; t6 := e5[13]; t7 := e5[4];
x := []; for y in [1..4] do x[y] := NullMat(4,4,Z(2)); od;
```

```

for y in [[1,1,3],[1,2,4],[2,1,2],[2,3,4],[3,2,1],[3,4,3],
          [4,3,1],[4,4,2]] do x[y[1]][y[2]][y[3]] := Z(2)^0; od;
h4 := NullMat(16,64,Z(2)); for y in [1..4] do
  z := KroneckerProduct(x[y],h0) + KroneckerProduct(h0,x[y]);
  h4 := h4 + KroneckerProduct(z,[h0[y]]); od;
I := Iterated([h4,h0,h0],KroneckerProduct);
H := f3(I,[2,3,0,1],[3,4,0,1,2]);
J := f3(I,[1,2,3,0],[1,2,3,4,0]);
B := KroneckerProduct(I,h0);
A := f3(B,[3,4,0,1,2],[4,5,0,1,2,3]);
C := f3(B,[2,3,4,0,1],[2,3,4,5,0,1]);
A := e2 * A * e1; B := e3 * B * e1; C := e4 * C * e1;
H := H * e3; I := I * e4; J := J * e2;
D := TransposedMat(A); K := TransposedMat(H);
F := TransposedMat(B); L := TransposedMat(I);
G := TransposedMat(C); M := TransposedMat(J);
x := A + A*G*L*I*C + A*G*C*F*B*D*A + A*G*C*D*A*F*B*D*A;
y := B + B*D*M*J*A + B*D*A*G*C*F*B + B*D*A*F*B*G*C*F*B;
z := C + C*F*K*H*B + C*F*B*D*A*G*C + C*F*B*G*C*D*A*G*C;
A := x; B := y; C := z; D := TransposedMat(A);
F := TransposedMat(B); G := TransposedMat(C);

```

8. Opal script p3.opal

```

let g = Graph([(1),(2),(3),(4),(5),(6),(7)],
  [A : (1) -> (2), B : (1) -> (3), C : (1) -> (4),
    D : (2) -> (1), F : (3) -> (1), G : (4) -> (1),
    H : (3) -> (7), I : (4) -> (5), J : (2) -> (6),
    K : (7) -> (3), L : (5) -> (4), M : (6) -> (2)]);
assume PathAlgebra(Integer(2),g);
let R = {D*A, F*B, G*C, K*H, L*I, M*J,
  D*B*H, F*C*I, G*A*J, K*F*A, L*G*B, M*D*C,
  K*F*C*G*B*H, L*G*A*D*C*I, M*D*B*F*A*J,
  D*C*G*A + D*B*F*A*J*M + J*M*D*B*F*A,
  F*A*D*B + F*C*G*B*H*K + H*K*F*C*G*B,
  G*B*F*C + G*A*D*C*I*L + I*L*G*A*D*C,
  D*B*F*C*G*A + D*C*G*B*F*A + D*B*F*A*D*B*F*C*G*A
  + D*B*F*A*D*C*I*L*G*A + D*B*F*C*G*A*D*B*F*A
  + D*C*G*A*D*B*F*C*G*A + D*C*G*B*H*K*F*C*G*A
  + D*C*I*L*G*A*D*B*F*A + D*B*F*A*D*C*G*B*F*C*G*A,
  F*C*G*A*D*B + F*A*D*C*G*B + F*C*G*B*F*C*G*A*D*B
  + F*C*G*B*F*A*J*M*D*B + F*C*G*A*D*B*F*C*G*B
  + F*A*D*B*F*C*G*A*D*B + F*A*D*C*I*L*G*A*D*B
  + F*A*J*M*D*B*F*C*G*B + F*C*G*B*F*A*D*C*G*A*D*B,
  G*A*D*B*F*C + G*B*F*A*D*C + G*A*D*C*G*A*D*B*F*C
  + G*A*D*C*G*B*H*K*F*C + G*A*D*B*F*C*G*A*D*C
  + G*B*F*C*G*A*D*B*F*C + G*B*F*A*J*M*D*B*F*C

```

```

+ G*B*H*K*F*C*G*A*D*C + G*A*D*C*G*B*F*A*D*B*F*C,
A*D*B*F*C*G + C*G*B*F*A*D + C*G*A*D*C*I*L*G*A*D*B*F*C*G*A*D
+ A*D*C*G*B*H*K*F*C*G + B*H*K*F*C*G*A*D*C*G
+ C*G*A*D*C*G*B*H*K*F + C*G*B*H*K*F*C*G*A*D
+ A*J*M*D*B*F*A*D*B*F*C*G + C*G*B*F*A*D*B*F*A*J*M*D
+ A*D*B*F*A*D*C*G*A*D*C*I*L*G + C*G*A*D*C*I*L*G*A*D*B*F*A*D
+ B*F*C*G*A*D + A*D*C*G*B*F + A*D*B*F*A*J*M*D*B*F*C*G*A*D*B*F
+ B*F*A*D*C*I*L*G*A*D + C*I*L*G*A*D*B*F*A*D
+ A*D*B*F*A*D*C*I*L*G + A*D*C*I*L*G*A*D*B*F
+ B*H*K*F*C*G*B*F*C*G*A*D + A*D*C*G*B*F*C*G*B*H*K*F
+ B*F*C*G*B*F*A*D*B*F*A*J*M*D + A*D*B*F*A*J*M*D*B*F*C*G*B*F,
B*F*C*G*A*D + A*D*C*G*B*F + A*D*B*F*A*J*M*D*B*F*C*G*A*D*B*F
+ B*F*A*D*C*I*L*G*A*D + C*I*L*G*A*D*B*F*A*D
+ A*D*B*F*A*D*C*I*L*G + A*D*C*I*L*G*A*D*B*F
+ B*H*K*F*C*G*B*F*C*G*A*D + A*D*C*G*B*F*C*G*B*H*K*F
+ B*F*C*G*B*F*A*D*B*F*A*J*M*D + A*D*B*F*A*J*M*D*B*F*C*G*B*F
+ C*G*A*D*B*F + B*F*A*D*C*G + B*F*C*G*B*H*K*F*C*G*A*D*B*F*C*G
+ C*G*B*F*A*J*M*D*B*F + A*J*M*D*B*F*C*G*B*F
+ B*F*C*G*B*F*A*J*M*D + B*F*A*J*M*D*B*F*C*G
+ C*I*L*G*A*D*C*G*A*D*B*F + B*F*A*D*C*G*A*D*C*I*L*G
+ C*G*A*D*C*G*B*F*C*G*B*H*K*F + B*F*C*G*B*H*K*F*C*G*A*D*C*G};
let S = Basis(R,600);
IsGrobner(S);
Nontips(S,18);

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Acknowledgement

I would like to thank Professor S. Koshitani for his valuable advice.

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